

STAT 412 QZ07 — Question 7

Problem

In a study of 799 randomly selected medical malpractice lawsuits, 509 were dropped or dismissed. Test the claim that most medical malpractice lawsuits are dropped or dismissed. Use the P-value method and the normal distribution as an approximation to the binomial distribution.

Solution

The word “most” means more than half, so

$$H_0 : p = 0.5, \quad H_a : p > 0.5.$$

This is choice B.

The sample proportion is

$$\hat{p} = \frac{509}{799} = 0.6370.$$

The one-proportion test statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.6370 - 0.5}{\sqrt{0.5(0.5)/799}} = 7.75.$$

This is a right-tailed test:

$$P = P(Z \geq 7.75) \approx 0.000.$$

Since the P-value is less than or equal to the significance level, reject H_0 .

Final Answer

Hypotheses: choice B. $z = 7.75$; P-value = 0.000. Conclusion about H_0 : choice B. Final conclusion: choice D.